

Dylan GIRAULT

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PROFILE

Software engineer (UTC 2026 diploma), with 3.5 years of experience in software development and testing within critical industrial environments.

SKILLS

- **Programming:** C++, Java, Python, C#, SQL, R, Go, JavaScript
- **Tools:** Jenkins, Qt, Git, Linux, Spring Boot, Rhapsody, AWS, Ranorex
- **Soft Skills:** Analytical mind, Adaptability, Teamwork, Precision
- **Languages:** French (Native), English (Advanced / C1)

PROFESSIONAL EXPERIENCE

Software Engineer - MBDA - Apprenticeship

September 2022 - February 2026

Le-Plessis-Robinson, France

- Implementation of an adaptable and efficient **testing solution** in **C++** and **JavaScript** within the department
- Development of **unit, integration and systems test validation**
- **50%** reduction in test development time
- Design of features in parallel with **tool support**

Research Engineer - University of Eastern Finland - Internship

February 2025 - May 2025

Joensuu, Finland

- Development of an **R** plugin for the **TNA (Transition Network Analysis) method** for the **Jamovi** tool
- Production of an article accepted by **ICSLE 2025** and winner of the **best article award**
- **92%** agreeing that the plugin adds value to the **TNA** (via the **TAM model**)

Test Analyst - EasyVista - Apprenticeship

September 2021 - August 2022

Noisy-le-Grand, France

- Implementation of automatic validation of a website, using **Ranorex** and **C#**
- Contribution to **continuous integration** through early detection of errors

ACADEMIC PROJECTS

Driving with Eyes - Autonomous Vehicle ROS2

- Team project (5 people): Full autonomous driving pipeline under **ROS2** on a Renault ZOE vehicle (perception, localization, navigation, control), deployed on an **embedded system** running **Linux**
- **Navigation Module Lead:** Determined the road segment to follow based on the vehicle's **pose** and steering **command**
- **Control Module Lead:** Implemented **lateral (PD)** and **longitudinal (PI)** control laws in real time, including a safety logic triggering vehicle stop upon loss of attention or absence of a valid command
- **Git Master:** Managed the repository and **Git workflow** for a team of 5 people on a **multi-module ROS2** project

Space Conquest - Multi-Agent Simulation

- **Multi-agent simulation in Go (backend) and React.js/TypeScript (frontend):** agent-countries compete to colonize planets by managing budget, strategies, and alliances
- Concurrent architecture: each agent runs in its own goroutine and communicates via **message-passing** to negotiate alliances
- Interactive **3D visualization** of the solar system with **React Three Fiber** to track the simulation in real time

ChatUTC - Real-Time Messaging Application

- Designed a **React/TypeScript** interface allowing users to create and join chat rooms, manage their profile, and invite other participants
- Implemented instant communication via **WebSocket**, **STOMP**, and **SockJS** with real-time display of messages and connected users
- Developed a secure API with **Spring Boot**, **Spring Security**, and **JWT**, with data persistence via **Spring Data JPA** and **MySQL**

Hot and Cold - Indoor Localization via iBeacon (Android)

- **Android** application in **Java** for locating an iBeacon indoors, in the style of a “hot or cold” game
- Used **Bluetooth Low Energy**, RSSI analysis, and triangulation calculations to estimate distance and direction
- Interface displaying a directional arrow and visual indicators to guide the user toward the target

Brick Breaker - Hybrid C / Assembly Architecture

- Mini-game in **C** with **GTK3** interface, pixel-by-pixel graphics rendering, and cross-implementation in **x86-64 assembly** to compare low-level and high-level approaches

EDUCATION

Computer Engineering degree - UTC (Université Technologique de Compiègne) Compiègne, France - Work-study program	2026
University Diploma in IT (DUT) - UPEC (Université Paris-Est Créteil) Fontainebleau, France - 2nd year, Work-study program	2022